

```
print("Smart City Traffic Forecasting Project Started")
```

Smart City Traffic Forecasting Project Started

```
from google.colab import files
uploaded = files.upload()
```

[Choose Files](#) train_aWnotuB.csv
train_aWnotuB.csv(text/csv) - 1815392 bytes, last modified: 5/27/2026 - 100% done
Saving train_aWnotuB.csv to train_aWnotuB (1).csv

```
import pandas as pd
import matplotlib.pyplot as plt
import os
```

```
os.listdir('/content')
```

```
['.config', 'train_aWnotuB (1).csv', 'train_aWnotuB.csv',  
'sample_data']
```

```
df = pd.read_csv('/content/train_aWnotuB.csv')
```

```
df.head()
```

	DateTime	Junction	Vehicles	ID	
0	2015-11-01 00:00:00	1	15	20151101001	
1	2015-11-01 01:00:00	1	13	20151101011	
2	2015-11-01 02:00:00	1	10	20151101021	
3	2015-11-01 03:00:00	1	7	20151101031	
4	2015-11-01 04:00:00	1	9	20151101041	

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

```
df.columns
```

```
Index(['DateTime', 'Junction', 'Vehicles', 'ID'], dtype='object')
```

```
df.shape
```

```
(48120, 4)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 48120 entries, 0 to 48119
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0   DateTime    48120 non-null  object
1   Junction    48120 non-null  int64
2   Vehicles    48120 non-null  int64
3   ID          48120 non-null  int64
dtypes: int64(3), object(1)
memory usage: 1.5+ MB
```

```
df.describe()
```

	Junction	Vehicles	ID	
count	48120.000000	48120.000000	4.812000e+04	
mean	2.180549	22.791334	2.016330e+10	
std	0.966955	20.750063	5.944854e+06	
min	1.000000	1.000000	2.015110e+10	
25%	1.000000	9.000000	2.016042e+10	
50%	2.000000	15.000000	2.016093e+10	
75%	3.000000	29.000000	2.017023e+10	
max	4.000000	180.000000	2.017063e+10	

```
df.isnull().sum()
```

	0
DateTime	0
Junction	0
Vehicles	0
ID	0

```
dtype: int64
```

```
df = df.dropna()
```

```
df.isnull().sum()
```

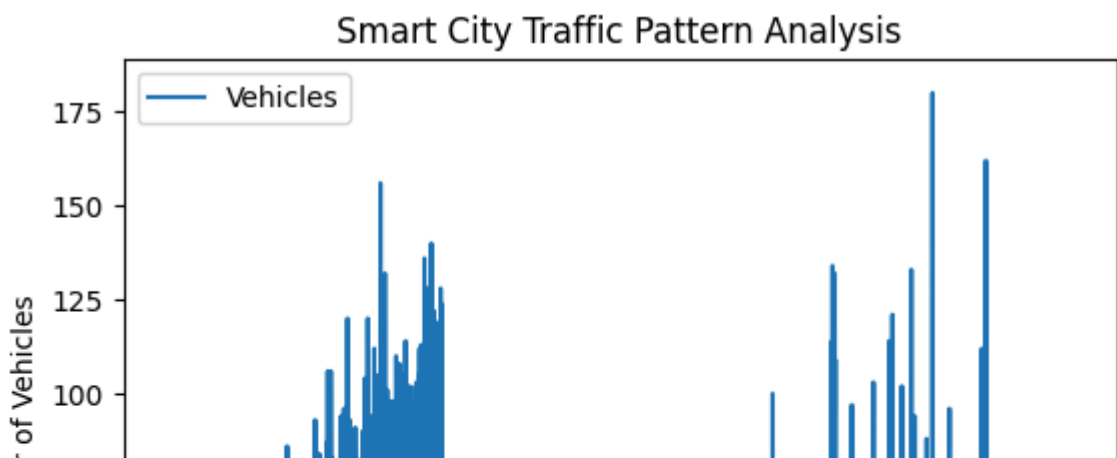
	0
DateTime	0
Junction	0
Vehicles	0
ID	0

dtype: int64

```
df.plot(x='DateTime', y='Vehicles', kind='line')
```

```
plt.title("Smart City Traffic Pattern Analysis")  
plt.xlabel("Date and Time")  
plt.ylabel("Number of Vehicles")
```

```
plt.show()
```



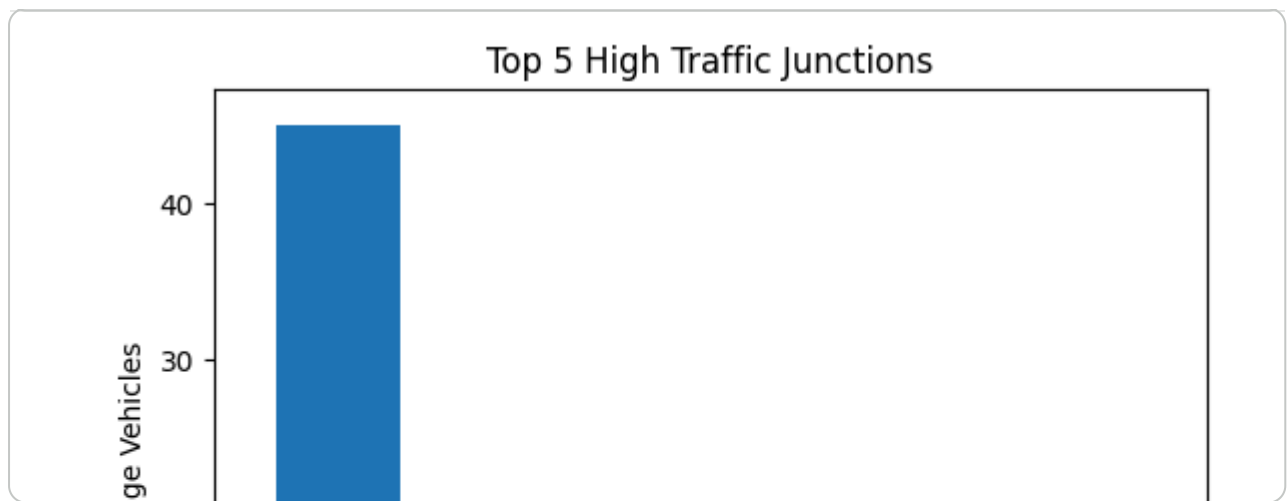
```
top5 = df.groupby('Junction')['Vehicles'].mean().sort_values(ascending=False)
```

```
top5.plot(kind='bar')
```

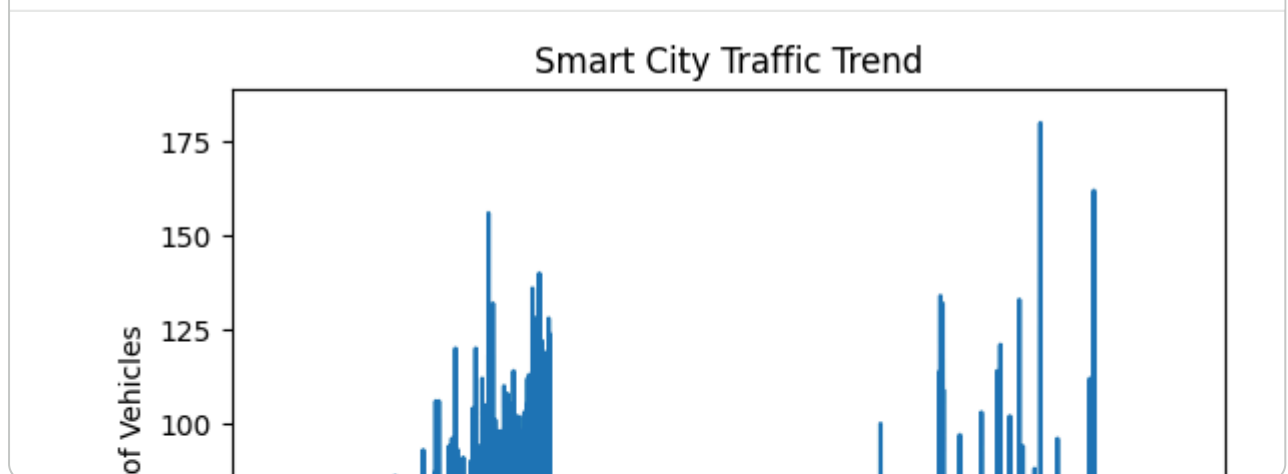
```
plt.title("Top 5 High Traffic Junctions")  
plt.xlabel("Junction")  
plt.ylabel("Average Vehicles")
```

```
plt.xticks(rotation=45)
```

```
plt.show()
```



```
traffic = df['Vehicles']  
  
plt.plot(traffic)  
  
plt.title("Smart City Traffic Trend")  
plt.xlabel("Time")  
plt.ylabel("Number of Vehicles")  
  
plt.show()
```



```
df.to_csv("cleaned_traffic_data.csv", index=False)
```

```
print("Smart City Traffic Forecasting Project Completed")
```

Smart City Traffic Forecasting Project Completed